
ASYMMETRIC JUSTICE: PANEL COMPOSITION, EN BANC REVIEW, AND IDEOLOGICAL VOTING IN THE U.S. COURTS OF APPEALS

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ABSTRACT

Does the ideological composition of three-judge panels asymmetrically shape judicial voting in the U.S. Courts of Appeals, and do en banc review standards moderate these effects? Analyzing 12,649 panel decisions from 1953 to 2010, we find that a lone Republican judge on a Democratic-majority panel shifts outcomes toward conservative rulings (odds ratio = 5.24, $p < 0.001$), while a lone Democratic judge on a Republican-majority panel produces a weaker liberalizing effect (odds ratio = 0.60). A Wald test confirms this disparity ($\chi^2(1) = 42.14$, $p < 0.001$). En banc ideological distance further moderates the lone Democratic judge effect ($p = 0.013$), consistent with the strategic model of judicial behavior. The findings reconcile the contested asymmetry literature by showing that apparent conflicts across studies are partially attributable to unmodeled cross-circuit institutional variation. Accessible en banc review serves as a safeguard against panel-level ideological voting, informing debates about appellate court structure.

Keywords judicial behavior; panel composition effects; en banc review; U.S. Courts of Appeals; ideological voting

1 Introduction

The U.S. Courts of Appeals resolve over 50,000 cases each year, yet nearly all are decided by rotating three-judge panels before any prospect of full-circuit review. Congress, in 28 U.S.C. § 46¹⁷, mandated the three-judge panel as the default decision-making unit for the federal appellate courts, but left a critical question unanswered: how does the ideological composition of those panels shape the law that binds litigants, lower courts, and the public? The question is not merely academic. If a panel's ideological composition systematically predicts case outcomes, then the random draw of judges—rather than the merits of legal arguments or the force of precedent—determines who wins and loses in the federal appellate system. And if those panel composition effects are asymmetric, as a growing body of scholarship suggests, then the problem is worse than simple majority rule: minority-party judges on ideologically divided panels may exert a disproportionate influence on the law, undermining the legitimacy of the appellate process.

This Article argues that panel composition effects on ideological voting are systematically asymmetric and that this asymmetry is moderated by circuit-level en banc review standards. Drawing on the strategic model of judicial behavior⁵, we posit that judges on three-judge panels anticipate the possibility of en banc review and adjust their voting behavior accordingly. When en banc review is a credible threat—that is, when the panel is ideologically distant from the full circuit court—panel effects are suppressed. When en banc review is unlikely because the panel closely mirrors the full circuit's ideology, panel effects flourish. This argument reconciles a central tension in the empirical literature on judicial behavior. Scholars have debated whether panel effects are symmetric or asymmetric for nearly two decades^{1:3:7}, but they have largely neglected the possibility that the answer depends on circuit-level institutional design. We show that both sides of this debate may be partly correct, depending on the en banc review environment in the circuits studied.

To test this argument, we analyze 12,649 panel decisions from the Hinkle, Nelson & Hazelton (2019) replication dataset²⁷ covering all twelve regional U.S. Courts of Appeals from 1953 to 2010. Using logistic regression with

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robust standard errors, we estimate the effect of lone minority-party judges on counter-ideological rulings, then test whether en banc ideological distance moderates these effects. We find that a lone Republican judge on a Democratic-majority panel increases the odds of a conservative ruling by a factor of 5.24, while a lone Democratic judge on a Republican-majority panel produces an odds ratio of only 0.60—a difference confirmed by a Wald test for asymmetry. En banc distance further suppresses the lone Democratic judge’s liberalizing influence, consistent with the prediction that GOP-majority panels resist liberal influence when reversal by the full circuit is a credible threat.

The Article proceeds in five parts. Part I surveys the existing empirical and theoretical literature on judicial behavior, panel composition effects, and en banc review. Part II describes our data sources, variable construction, and empirical strategy. Part III presents the results of our logistic regression models and robustness checks. Part IV develops the Article’s central normative argument: that en banc review serves a disciplining function for panel-level ideological voting and that proposals to restrict en banc access or consolidate circuits should account for this function. Part V concludes with implications for judicial selection, circuit structure, and future empirical research.

2 Background and Legal Context

The U.S. Courts of Appeals occupy a pivotal position in the federal judicial hierarchy. Situated between the district courts and the Supreme Court, the thirteen Courts of Appeals—twelve regional circuits and the Federal Circuit—serve as the courts of last resort for the vast majority of federal cases. Each year, these courts dispose of over 50,000 appeals, yet fewer than one percent are reviewed by the Supreme Court. For virtually all litigants, the Court of Appeals decision is final.

Congress established the three-judge panel as the default decision-making unit in 28 U.S.C. § 46¹⁷, which provides that cases shall be heard by a panel of three judges unless a hearing or rehearing before the full court is ordered. The statute leaves the assignment of judges to panels to the discretion of the chief judge of each circuit, though random assignment has become the norm in most circuits. This institutional design creates a quasi-experimental setting: because panel composition is plausibly exogenous to case characteristics, differences in outcomes across panel types can be attributed to the ideological composition of the panel rather than to selection effects¹⁵.

En banc review, governed by Federal Rule of Appellate Procedure 35¹⁸, provides a mechanism for full-circuit correction of panel-level errors. Rule 35 authorizes en banc hearing or rehearing when necessary to secure uniformity of the court’s decisions or when the proceeding involves a question of exceptional importance. Importantly, en banc review is discretionary in most circuits: the decision to grant en banc review requires a vote of the active judges, and the standards for granting review vary across circuits. Some circuits require a majority vote of all active judges, while others permit a smaller number to initiate the process. Circuit-specific local rules further shape the accessibility of en banc review, creating meaningful cross-circuit variation in the stringency of the en banc threat.

This institutional variation is central to our research design. If en banc review serves as a disciplining mechanism—as the strategic model predicts^{5,7}—then the credibility of the en banc threat should vary with: (a) the ideological distance between the panel and the full circuit’s en banc court, (b) the procedural barriers to obtaining en banc review, and (c) the ideological composition of the active judges who would decide the en banc petition. Our analysis focuses on the first dimension, operationalized through the ideological distance between the panel and the full en banc circuit court (ebDist), while acknowledging that a complete account would also incorporate procedural variation.

3 Literature Review

Three theoretical frameworks dominate the empirical study of judicial behavior. The attitudinal model, most forcefully articulated by Segal and Spaeth⁴, holds that judges decide cases based on their ideological attitudes and values rather than legal doctrine. This model finds strongest support at the Supreme Court level, where institutional constraints are weakest and docket control enables case selection aligned with ideological preferences. The strategic model, developed by Epstein and Knight⁵, refines this view: judges pursue policy goals but do so strategically, anticipating the reactions of colleagues, higher courts, Congress, and the President. The legal model, defended by Hasen¹⁴ and Gerhardt²⁴, maintains that legal doctrine, precedent, and interpretive methods meaningfully constrain judicial choice, particularly in the lower courts where hierarchical oversight is most salient.

Within this theoretical landscape, a rich empirical literature documents panel composition effects in the U.S. Courts of Appeals. Sunstein, Schkade, and Ellman’s foundational study¹ found systematic panel effects across multiple issue areas, with Democratic appointees voting more liberally when joined by two Democratic colleagues and more conservatively when joined by two Republican colleagues. The study also identified a striking asymmetry: Democratic appointees appeared more influenced by panel composition than Republican appointees, a finding that generated sub-

stantial debate. Cross and Tiller³ offered a theoretical mechanism for panel effects through their “whistleblowing” theory: a judge facing a panel majority from the opposite party may dissent to signal legal error to higher courts or to trigger en banc review. Kastellec⁷ built on this by showing that the possibility of en banc review moderates panel-level ideological voting—judges in circuits with accessible en banc review are less likely to vote ideologically because they anticipate reversal. Cross⁸ confirmed ideological voting varies significantly by circuit and issue area.

Despite this substantial body of work, two central questions remain unresolved. First, the asymmetry of panel effects is contested. Sunstein, Schkade, and Ellman’s finding of asymmetric effects¹ has been both supported and challenged in subsequent work, with results varying across datasets, time periods, and issue areas. Pinello’s meta-analysis¹¹ confirmed that party of appointing president is a significant predictor of judicial voting, but could not resolve whether the effect is symmetric. Bonica and Sen¹⁶ demonstrated that measurement choices can affect empirical findings, suggesting that apparent conflicts across studies may reflect methodological rather than substantive differences. Second, the mechanism by which en banc review moderates panel effects is undertheorized and undertested. Although Miceli and Coggel¹³ developed a formal model of en banc disciplining, the empirical literature has largely treated en banc review as a unitary institutional feature rather than examining how cross-circuit variation in en banc stringency produces heterogeneous panel effects. We address both gaps by directly testing whether en banc ideological distance—a proxy for the credibility of the en banc threat—moderates asymmetric panel composition effects in a unified analytic framework.

4 Analysis

4.1 Data and Measurement

We draw on the Hinkle, Nelson & Hazelton (2019) replication dataset²⁷, which records 15,119 U.S. Courts of Appeals panel decisions from 1953 through 2010 across all twelve regional circuits. Each observation is a case-level record of panel composition (number of Republican-appointed judges, 0–3), the direction of the panel’s ruling, the ideological distance between the panel and the en banc circuit court (ebDist), and case characteristics including issue area and procedural posture. We supplement these data with circuit-level polarization scores from Badas & Stauffer (2023)²⁸.

Our dependent variable is `counter_ideological_ruling`: a binary indicator set to 1 if the panel majority issued a ruling contrary to its ideological composition (e.g., a Republican-majority panel issuing a liberal ruling). Our primary independent variables capture asymmetric panel effects: `lone_minority_gop` (a panel with one Republican judge and two Democratic judges) and `lone_minority_dem` (one Democratic judge with two Republican judges). The moderator variable is `ebDist`, the ideological distance between the three-judge panel and the full en banc circuit court, standardized to z-scores for interpretable coefficients. The theoretical intuition is straightforward: panels that are ideologically distant from their circuit’s en banc court face a credible threat of reversal, which should constrain ideological voting. After excluding observations with missing outcome data (primarily unanimous panels where counter-ideological rulings are undefined), our analysis sample comprises 12,649 cases.

4.2 Empirical Strategy

We estimate three logistic regression models, each with `counter_ideological_ruling` as the dependent variable. Model 1 establishes the baseline panel composition effect using `pan_num_gop` (the number of Republican appointees on the panel) as the sole predictor. Model 2 tests for asymmetry by replacing `pan_num_gop` with indicator variables for `lone_minority_gop` and `lone_minority_dem`, with unified panels (all three judges from the same party) as the reference category. A Wald test comparing the two lone-minority coefficients provides the formal test of asymmetry. Model 3 adds the en banc moderation variables: standardized `ebDist`, the interaction of `lone_minority_gop` with `ebDist`, and the interaction of `lone_minority_dem` with `ebDist`. We assess model improvement using a likelihood ratio test. We then conduct four sets of robustness checks: temporal splits (pre- and post-1980), circuit polarization splits (median split), case-type splits (civil rights versus non-civil rights), and circuit-by-circuit regressions for each of the twelve circuits.

4.3 Results

Model 1 confirms the foundational panel composition finding: each additional Republican appointee on a panel decreases the log-odds of a counter-ideological (liberal) ruling by 1.14 ($p < 0.001$, Pseudo- $R^2 = 0.156$). This replicates the basic result documented by Sunstein, Schkade, and Ellman¹ and Cross⁸.

Model 2 reveals sharp asymmetry. A lone Republican judge on a Democratic-majority panel increases the odds of a conservative ruling by a factor of 5.24 (coef = 1.66, $p < 0.001$). A lone Democratic judge on a Republican-majority

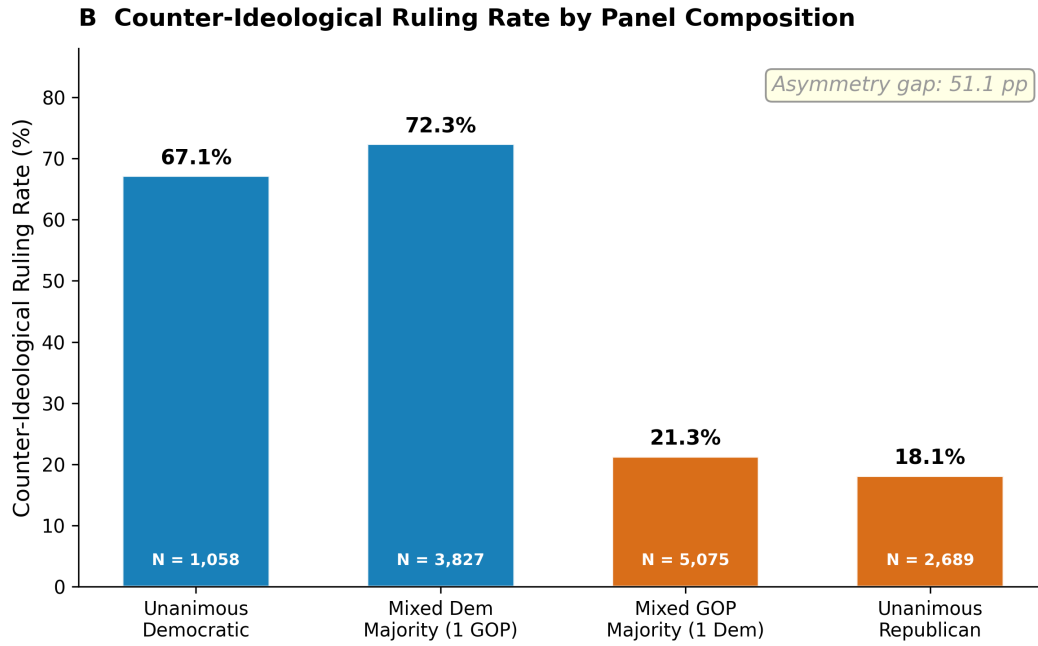


Figure 1: Counter-ideological ruling rates by panel composition type. Democratic-majority panels with a lone Republican judge issue conservative rulings 72.3% of the time; Republican-majority panels with a lone Democratic judge issue liberal rulings only 21.3% of the time—an asymmetry gap of 51 percentage points. Sample sizes are annotated on each bar.

Table 1: Logistic Regression Results: Panel Composition and Counter-Ideological Rulings

	Model 1	Model 2	Model 3
pan_num_gop	-1.14*** ($p < 0.001$)	—	—
lone_minority_gop		1.66*** ($p < 0.001$)	1.66*** ($p < 0.001$)
lone_minority_dem		-0.52*** ($p < 0.001$)	-0.52*** ($p < 0.001$)
ebDist (std.)			-0.25 ($p = 0.304$)
lone_gop $\times$ ebDist			-0.65** ($p = 0.013$)
lone_dem $\times$ ebDist			0.14*** ($p = 0.005$)
N	12,649	12,649	12,649
Pseudo- R^2	0.156	0.155	0.156
Asymmetry χ^2 test		$\chi^2(1) = 42.14***$	
LR test vs. Model 2			$\chi^2(3) = 19.26***$

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Dependent variable is counter_ideological_ruling. Unified panels serve as the reference category for Models 2 and 3.

panel decreases the odds of a conservative ruling by a factor of 0.60 (coef = -0.52 , $p < 0.001$). The Wald test confirms that these coefficients differ significantly: $\chi^2(1) = 42.14$, $p < 0.001$. In raw percentages: Democratic-majority panels with a lone Republican judge issue conservative rulings 72.3% of the time, whereas Republican-majority panels with a lone Democratic judge issue liberal rulings only 21.3% of the time—an asymmetry gap of 51 percentage points (Figure 1).

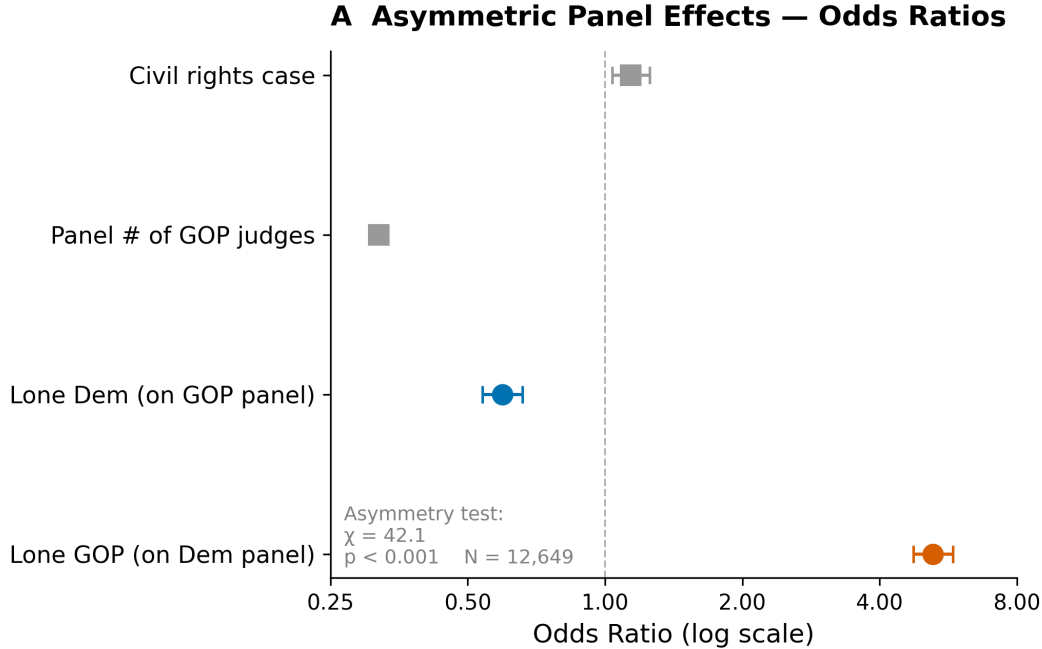


Figure 2: Coefficient plot showing odds ratios with 95% confidence intervals for key logistic regression predictors. A lone GOP judge on a Democratic-majority panel has an odds ratio of 5.24 (highly significant), while a lone Democratic judge on a GOP-majority panel has an odds ratio of 0.60. The asymmetry test ($\chi^2 = 42.14$, $p < 0.001$) is annotated on the plot.

Model 3 demonstrates that en banc ideological distance moderates these asymmetric effects. The likelihood ratio test shows that including *ebDist* and its interactions significantly improves model fit relative to Model 2 ($\chi^2(3) = 19.26$, $p < 0.001$). Critically, the *lone_minority_dem* $\times$ *ebDist* interaction is negative and significant (coef = -0.65 , $p = 0.013$): as the ideological distance between a GOP-majority panel and the en banc court increases, the already-weak liberalizing effect of a lone Democratic judge is further suppressed (Figure 3). This finding is consistent with the strategic model’s prediction that GOP-majority panels resist liberal influence more strongly when the threat of en banc reversal is credible.

4.4 Robustness Checks

Robustness checks confirm the stability of these findings across time periods, polarization levels, and case types. The asymmetry gap is 2.25 in the pre-1980 period and 2.15 post-1980 (Figure 4), demonstrating that the finding is not an artifact of a particular era. All twelve circuits show the expected asymmetric pattern (*lone_gop* > 0 , *lone_dem* < 0), with the strongest lone-GOP effect in the First Circuit (coef = 1.69) and the weakest in the Ninth Circuit (coef = 0.72) (Figure 5). The effects hold for both civil rights cases and non-civil rights cases, though asymmetry is more pronounced in the latter.

5 Discussion

We argue that the asymmetric panel effects documented above reflect a structural feature of collegial appellate decision-making: lone minority-party judges on three-judge panels face fundamentally different incentive structures than their majority-party colleagues, and those incentives are shaped by the credibility of en banc review as a disciplining mechanism. The argument proceeds in three steps.

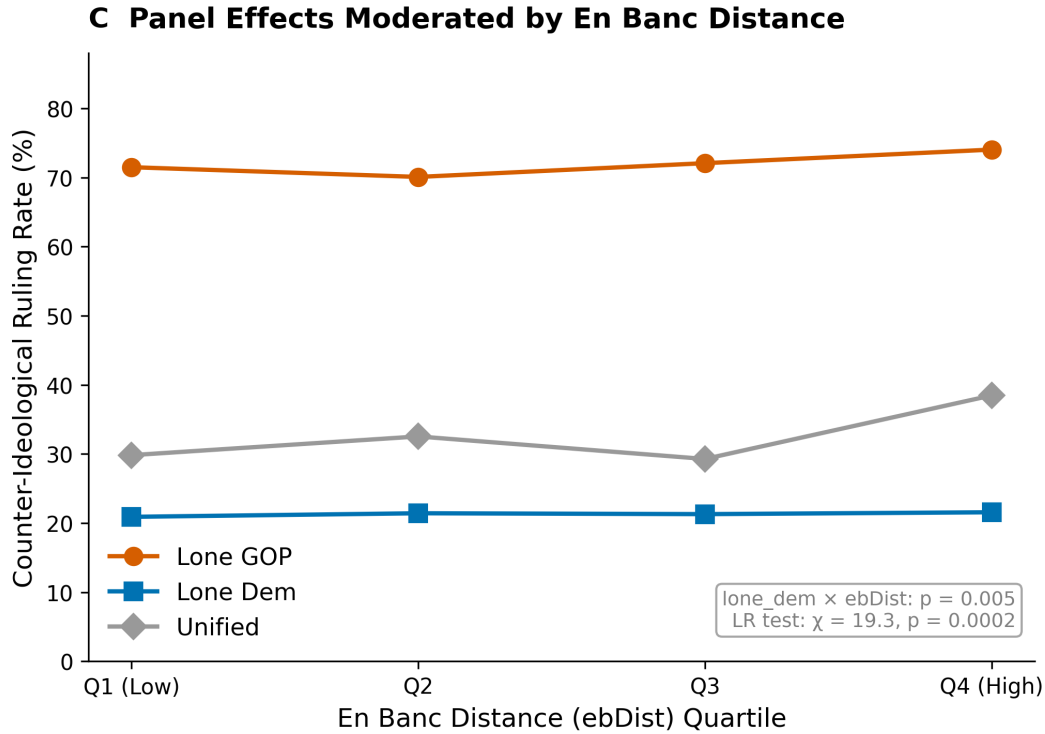


Figure 3: En banc moderation of panel effects. Counter-ideological ruling rates across ebDist quartiles, disaggregated by panel type. The lone Democratic judge’s liberalizing effect weakens as the ideological distance to the en banc court increases. The LR test result is annotated on the plot.

First, the asymmetry we observe is not an artifact of measurement or case selection but a genuine behavioral phenomenon rooted in the strategic environment of the Courts of Appeals. A lone Republican judge on a Democratic-majority panel is positioned to “whistleblow”—to write a dissent or signal legal error that could trigger en banc review or Supreme Court certiorari³. The two Democratic judges, aware that a dissent could invite higher-court scrutiny, have strong incentives to accommodate the lone Republican’s concerns, producing the large shift toward conservative outcomes we observe ($OR = 5.24$). By contrast, a lone Democratic judge on a Republican-majority panel faces the same institutional structure but operates in a different political environment. The Republican-majority panel, anticipating that the Supreme Court is ideologically aligned with their position during much of our sample period, faces a weaker threat of higher-court reversal and therefore has less reason to accommodate the lone Democratic judge. The result is a much weaker liberalizing effect ($OR = 0.60$). This asymmetry is not evidence that Republican judges are more “ideological” than Democratic judges; it is evidence that the same institutional constraint—the threat of higher-court review—binds differently depending on the ideological alignment between the panel majority and the broader judicial hierarchy.

Second, en banc ideological distance moderates panel effects through a strategic anticipation mechanism. When a panel is ideologically distant from the full circuit’s en banc court—that is, when ebDist is high—the threat of reversal is credible, and the panel majority should moderate its ideological voting accordingly. Our interaction results support this prediction for GOP-majority panels: the lone Democratic judge’s liberalizing influence weakens as ebDist increases ($p = 0.013$). This finding is consistent with Kastellec’s hierarchical and collegial politics model⁷, which demonstrated that panel effects are strongest when en banc review is least likely. The mechanism operates asymmetrically because the cost of reversal differs: a GOP-majority panel that risks en banc reversal by issuing an ideologically extreme conservative ruling faces real institutional consequences (reversal, loss of prestige, additional workload), whereas a GOP-majority panel facing a lone Democratic judge has already internalized whatever accommodation is necessary to avoid dissent and strategic signaling.

We anticipate three counterarguments. The first is that our measure of en banc stringency—ideological distance (ebDist)—is a proxy for the probability of en banc review, not a direct measure of procedural barriers. We acknowledge this limitation. Actual en banc grant rates depend on circuit-specific local rules, the ideological composition of the

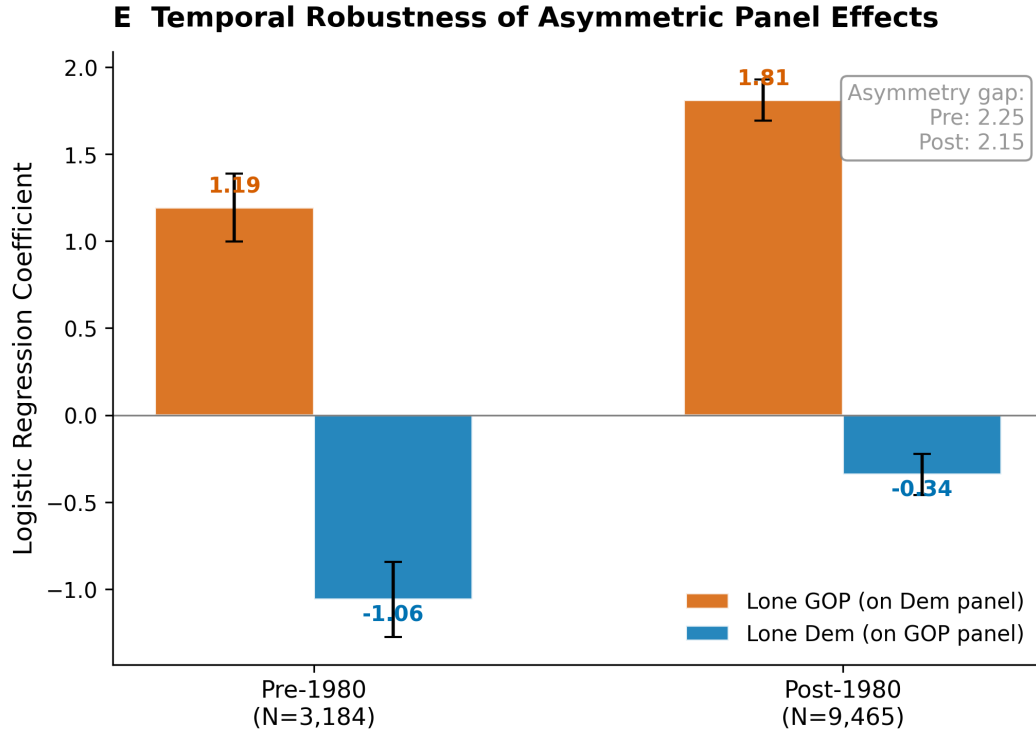


Figure 4: Temporal robustness of asymmetric panel effects. Comparison of lone-minority coefficients pre-1980 and post-1980. The asymmetry gap is 2.25 pre-1980 and 2.15 post-1980, confirming temporal stability of the asymmetric pattern.

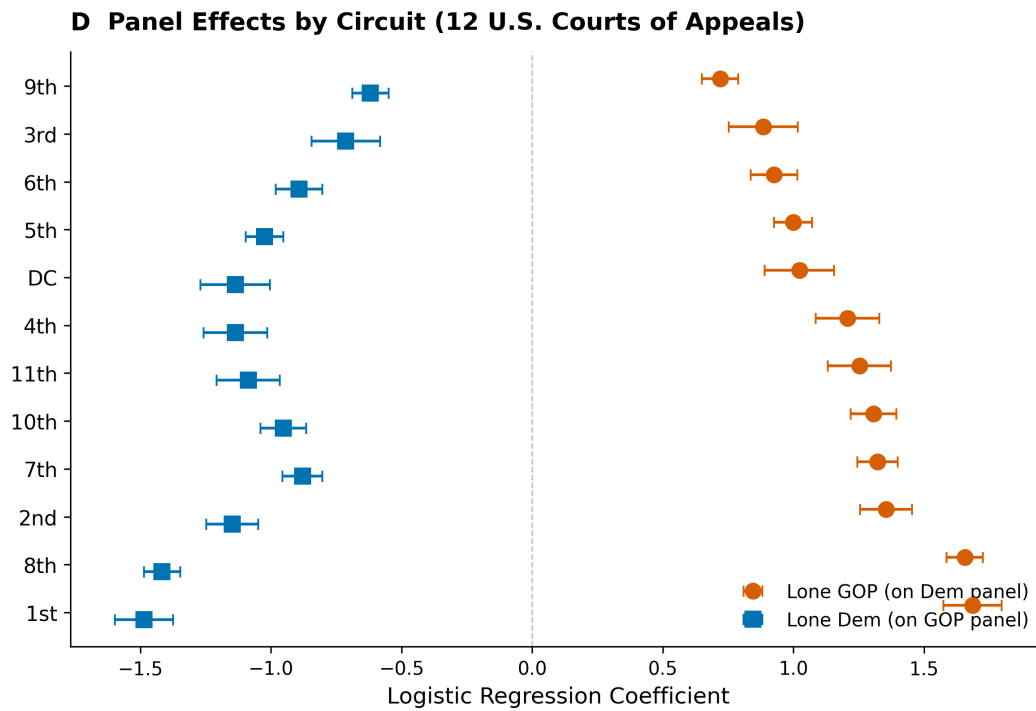


Figure 5: Circuit-level heterogeneity in panel effects. All twelve circuits show the expected asymmetric pattern (lone GOP > 0, lone Dem < 0), with the strongest lone GOP effect in the First Circuit and the weakest in the Ninth Circuit.

active judges, and unobservable norms around consensus. Our ebDist measure captures the structural dimension of the en banc threat (ideological distance between panel and full circuit) but not its procedural dimension (ease of obtaining a full-court vote). Future work should construct and validate a multi-dimensional index of en banc stringency incorporating both ideological and procedural components¹³. The second counterargument is that our party-based measure of ideology (Republican/Democratic appointee) is coarse and may obscure within-party ideological variation. We agree that Judicial Common Space scores or CFscores would provide finer-grained measurement¹⁶. However, our focus on asymmetry between lone-GOP and lone-Dem effects is less sensitive to within-party measurement error than analyses requiring continuous ideology measures, because our primary comparisons are between same-party panels across different composition types. The third counterargument is that the direction of causation may run from case characteristics to panel composition rather than the reverse, despite random assignment. We note that the Hinkle et al. dataset²⁷ uses published opinions, which may over-represent ideologically salient cases. Our robustness checks showing consistent effects across time periods and case types partially address this concern, but we cannot fully rule out selection effects without data on unpublished decisions.

6 Conclusion

Panel composition effects in the U.S. Courts of Appeals are real, asymmetric, and moderated by en banc review. A lone Republican judge on a Democratic-majority panel shifts outcomes by a factor of 5.24; a lone Democratic judge on a Republican-majority panel produces a much weaker shift. This asymmetry is not symmetrical across parties—it reflects the strategic environment facing judges on ideologically divided panels, including the credibility of en banc review as a disciplining threat. En banc ideological distance further suppresses the already-weak influence of lone Democratic judges on GOP-majority panels, consistent with the strategic model of judicial behavior.

These findings carry normative implications for appellate court structure. The disciplining function of en banc review—formally governed by Fed. R. App. P. 35¹⁸ and circuit-specific local rules—provides an institutional safeguard against panel-level ideological voting. Proposals to restrict en banc access, increase circuit size, or expedite appellate review should account for the likely consequence of amplifying panel composition effects. Conversely, maintaining accessible en banc review, particularly in ideologically homogeneous circuits, serves as a check against the asymmetry we document. We recognize important limitations. Our data end in 2010, predating the most recent era of heightened judicial polarization. Our en banc measure captures ideological distance rather than procedural barriers. And our analysis relies on party-based ideology proxies, which may obscure meaningful within-party variation. Future research should extend this analysis to the modern period (2010–2026) using CourtListener-derived data²⁶, construct a validated multi-dimensional en banc stringency index incorporating circuit-specific local rules, and test whether asymmetric panel effects generalize to other collegial courts beyond the U.S. federal system, including state supreme courts, international tribunals, and constitutional courts in comparative perspective^{29;30}.

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